Answer key: Lesson 8, Graph Detectives

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Study the graph. Each picture stands for 1 book. Answer questions 1 to 3.

Answer: Mei; Siti

Mei has 9 pictures. That is the longest row, so Mei read the most books. Ali has 6 pictures. Siti also has 6, so Ali and Siti read the same number of books.

Watch for: Choosing Ali because his name is first.; Writing 6 instead of a name.

S2. Look at the graph in question 1.

Answer: 5 more books; 25 books in all

Mei 9 and Ben 4. 9 take away 4 is 5 more books. All four rows: 6 + 9 + 4 + 6 = 25 books.

Watch for: Adding 9 and 4 to find how many more.; Leaving out a row when finding the total.

S3. Look at the graph in question 1.

Answer: True, True, False, False

Ben has 4 pictures, the fewest: true. Mei 9 and Ali 6: 9 take away 6 is 3 more: true. Ali 6 and Siti 6 are the same, so Ali did not read more: false. 6 + 9 + 4 + 6 = 25, not 24: false.

Watch for: Saying Ali read more than Siti because Ali is first.; Adding only three rows to find the total.

S4. Study the graph. Each picture stands for 1 fruit. Answer questions 4 to 6.

Answer: 4 kinds; Oranges

There are 4 rows: apples, pears, plums and oranges. Pears have 5 pictures and oranges have 5 pictures: the same number.

Watch for: Counting the pictures instead of the rows.; Choosing plums because it is next to pears.

S5. Look at the graph in question 4.

Answer: 2 fewer plums than pears; 11 fruits

Pears 5 and plums 3. 5 take away 3 is 2 fewer plums. Apples 8 and plums 3: 8 + 3 = 11 fruits.

Watch for: Adding 5 and 3 to find how many fewer.; Adding the pears as well.

S6. Look at the graph in question 4.

Answer: 21 fruits in all; Apples

8 + 5 + 3 + 5 = 21 fruits. Apples have 8 pictures, the longest row, so there are the most apples.

Watch for: Leaving out a row when finding the total.; Choosing the kind with the most letters.

Think harder (page 3)

T1. There are 18 fish in all. There are 2 fewer blue fish than red fish. Draw the blue fish and the gold fish in the empty boxes.

Answer: Blue 5; gold 6

Blue: 7 take away 2 is 5. Red and blue make 7 + 5 = 12. 18 take away 12 is 6 gold fish. Check: 7 + 5 + 6 = 18.

Watch for: Writing 2 for the blue fish (the number fewer, not the number of fish).; Drawing a picture in every empty box.

T2. Study the graph. Each picture stands for 2 shells. Answer questions T2 and T3.

Answer: Pearl 6 shells; 4 more shells

Pearl has 3 pictures: 2, 4, 6 is 6 shells. Reef has 5 pictures: 2, 4, 6, 8, 10 is 10 shells. Reef has 2 more pictures than Pearl, and 2 pictures are 4 shells, so Reef has 4 more shells.

Watch for: Writing 3 for the shells in Pearl's row (counting pictures).; Writing 2 for how many more shells (2 more pictures).

T3. A child in another class looks at the graph in question T2. She says Coral has 4 shells.

Answer: No; 8 shells

The key says each picture stands for 2 shells. Coral has 4 pictures: 2, 4, 6, 8. Coral has 8 shells, not 4.

Watch for: Agreeing because she counted 4 pictures.; Writing 4 for the number of shells.

Go further (page 4)

G1. Mei's graph uses a key. Each picture stands for 3 stickers. There are 24 stickers in all. Draw the pictures for Lily in the empty boxes.

Answer: 15 stickers; 3 pictures

Mei has 3 pictures: 3, 6, 9 is 9 stickers. Ravi has 2 pictures: 3, 6 is 6 stickers. 9 + 6 = 15. 24 take away 15 is 9 stickers for Lily: 3, 6, 9 is 3 pictures. Check: 8 pictures of 3 are 24.

Watch for: Writing 9 pictures for Lily (the number of stickers, not pictures).; Counting the pictures as 1 sticker each.

G2. A graph shows 18 birds in all. There are the same number of ducks and geese. There are 3 more hens than ducks.

Answer: Ducks 5; hens 8

Try 4 ducks: 4 geese and 7 hens make 15, too few. Try 5 ducks: 5 geese and 8 hens make 18. So ducks 5, geese 5, hens 8. Check: 5 + 5 + 8 = 18.

Watch for: Writing 3 for the hens (the number more, not the number of hens).; Forgetting that the geese are the same as the ducks.

G3. Each picture stands for 1 book. Which question can the graph not answer? Write its number in the brackets.

Answer: (2)

The graph shows how many books of each kind there are. Question (2) asks which book is the best: the graph does not show that. The other three can be answered: 9 story books; 5 more story books than art books; 24 books in all.

Watch for: Choosing (3) because it needs a sum.; Choosing a question that has a number in it.

Investigation (page 5): Look, count and ask

Many answers are right. A good graph has a title, the kinds named, one picture for each thing, the same size, and a key. A question it can answer is about the numbers (how many, how many more); a question it cannot answer is about likes, feelings or reasons.

Look for: A child who counts three kinds, draws one picture for each, and writes one question about the numbers and one about likes or reasons that the graph cannot answer. Credit any clear pair. A child who writes two questions the graph can answer has not yet met the idea: ask, what does the graph not show? A child who uses a key of 2 has gone further.

Exit ticket (page 5)

E1. Look at the graph. Each picture stands for 1 sticker.

Answer: Siti

Siti has 3 pictures. That is the shortest row, so Siti has the fewest stickers.

Watch for: Choosing the longest row.; Choosing the first name.

E2. Look at the graph in E1.

Answer: 16 stickers

Ali 5, Mei 8 and Siti 3. 5 + 8 + 3 = 16 stickers.

Watch for: Leaving out a row.; Writing the number of rows.

E3. Look at the graph in E1. Mei has 3 more stickers than Ali.

Answer: True

Mei 8 and Ali 5. 8 take away 5 is 3 more: true.

Watch for: Saying false because Ali comes first.; Adding 8 and 5.

Sneak-peek puzzles (not in the workbook)

1. 3 pictures.
2. 6 stickers. Each picture stands for 2, so count in 2s: 2, 4, 6.